

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0545

Mount View 138 kV
January 16, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0545. The Interconnection Customer proposes a 69.5 MW solar photovoltaic facility interconnecting at the Mount View 138 kV substation in the Virginia area. The request was received on December 03, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 277 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,531,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Rustic to Hughson Substation 345 KV
Contingency: Dixleta to Rustic 345 KV
Loading: 127.5% (Rating: 1,970 MVA)
DFAX: 7.0%
Pre-project loading: 127.2%
2. Facility: Bixby Hill to Tombstone 500 KV
Contingency: Tombstone to Sharyland Substation 345 KV
Loading: 117.8% (Rating: 3,937 MVA)
DFAX: 5.9%
Pre-project loading: 117.7%
3. Facility: Louisa Generation Facility to Carrabelle 345 KV
Contingency: N Anna to Louisa Generation Facility 345 KV
Loading: 115.1% (Rating: 1,907 MVA)
DFAX: 5.4%
Pre-project loading: 114.9%
4. Facility: Tom Watkins to Kimper 138 KV
Contingency: Atoka to Catawba 345 KV L
Loading: 112.8% (Rating: 305 MVA)
DFAX: 6.2%
Pre-project loading: 111.4%
5. Facility: Leavitt Substation to Birchwood 345 KV
Contingency: Bear Island to Dunn Siding 345 KV
Loading: 107.6% (Rating: 2,449 MVA)
DFAX: 5.4%
Pre-project loading: 107.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Rustic to Hughson Substation 345 KV	Rebuild line with higher-capacity conductor	\$53,446,000	1% (\$574,000)
Bixby Hill to Tombstone 500 KV L	Rebuild line with higher-capacity conductor	\$162,073,000	1% (\$1,793,000)
Louisa Generation Facility to Carrabelle 345 KV	Rebuild line with higher-capacity conductor	\$107,381,000	1% (\$788,000)
Tom Watkins to Kimper 138 KV	Rebuild line with higher-capacity conductor	\$6,489,000	1% (\$90,000)
Leavitt to Birchwood 345 KV L	Reconductor line	\$38,643,000	1% (\$286,000)
	Total allocated to project		\$3,531,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.